

Karma

truth & existing positions in the prediction market.
bull coin holder already believes $\rightarrow$ price will go up.
bear coin $\rightarrow$ price will go down.

Self oracking $\rightarrow$ participants' economic exposure
already encodes their beliefs.
Reporters are the market.

Formal Model -

Participants: set $P = \{1, 2, \dots, n\}$ of token holders.
state for participant i @ time t :

$B_i(t) \rightarrow$ Bull coin balance

$b_i(t) \rightarrow$ Bear coin balance

$\mathbf{B}_i(t) = (B_i(t), b_i(t))$

Price submission - each participant submits
 $P_i(t) \in \mathbb{R}^+$

Weighted func. $= W_i(t) = \sqrt{B_i(t) \times b_i(t)} \times D(t - t_{\text{last}}^i)$

where D is a decay func. (from 0 to 1) that $\downarrow$ w/ if
participant hasn't submitted recently.

$$P_{\text{agg}}(t) = \frac{\sum W_i(t) \cdot P_i(t)}{\sum W_i(t)}$$

Theorem 1 - [wt func. Optimality]

for a participant with balances (B, b) , any $W(B, b)$
that satisfies - (a) $W(B, 0) = W(0, b) = 0$ [boundary]

$\downarrow$ (b) $W(B, b) = W(b, B)$ [symmetry]
must be (c) $W(\lambda B, \lambda b) = \lambda W(B, b) \in \lambda > 0$ [scale invariance]

of the form (d) $\frac{\partial^2 W}{\partial B^2} < 0, \frac{\partial^2 W}{\partial b^2} < 0$ [concave]

$\downarrow$ (e) $\frac{\partial W}{\partial B} > 0, \frac{\partial W}{\partial b} > 0$ [monotonicity]

for $\alpha \in (0, 1)$. Geometric mean @ $d = 0.5$.

Proof - Scale invariance requires W to be homogeneous of
degree 1. Euler's homogeneous func. -

$$W(B, b) = B \frac{\partial W}{\partial B} + b \frac{\partial W}{\partial b}$$

symmetry $\rightarrow$ marginal contributions are equal @ $B=b$
i.e. $\frac{\partial W}{\partial B} = \frac{\partial W}{\partial b}$ @ $B=b$. Concavity restricts the

func. form to one with linearity returns each arg.
is homogeneous - Cobb-Douglas form.

$W(B, b) = C B^\alpha b^{1-\alpha}$. Geometric mean
emerges from additional req. that W is max for a
fixed sum $B+b$ when $B=b$.

Participant's Utility func. -
each participant maximizes expected
utility over time.

$U_i = E \left[\sum_{t=0}^{\infty} \delta^t u_i(t) \right]$ where $\delta \in (0, 1)$
(discount factor)
 $u_i(t)$ has 2 components: trading profit / patience

1. Trading profit -
 $\pi_i^{\text{trade}}(t) = [P_i^{\text{submit}}(t) - P_{\text{agg}}(t)] \cdot \frac{B_i(t)}{B_i(t)}$

If net bull $\rightarrow B_i > b_i \rightarrow$ profit from high
submissions. net bear $\rightarrow$ profit from low.

2. wt-based profit -

$$R_i(t) = \frac{W_i(t)}{\sum W_j(t)} \cdot \text{total_reward}(t)$$

rewards (trade fees, inflation) & wt.

total utility $\rightarrow U_i(t) = \pi_i^{\text{trade}}(t) + R_i(t)$

Trade Offs -

conflicting objectives -

short term - manipulate π_i^{trade} to $\uparrow$
 π_i^{trade}

long term - manipulate W_i to $\uparrow$
future R_i .

[short term - participants can manipulate submit
price to boost immediate payoff, short-lived
trading advantages.]

[long term - success (sustained) depends on high
wt W_i ; only maintained. W depends on balances
& submission activity with decay. Determines
how much influence a participant has in
future reward allocation R_i . Consistently
keeping W_i strong = larger rewards.]

Basically, when a biased price submitted,
if bull holder ($B_i > b_i$) submit high, temp.
profit but this affects aggregated price which
affects market value of both tokens which
changes (B_i, b_i) pos $u_i(t)$ future gets affected.
Feedback loop.

General Cobb-Douglas form:

$$W_{\text{neutrality}}(B, b) = B^\alpha b^{1-\alpha} \quad \left[\begin{array}{l} \text{Karma specific} \\ \text{in M form} \end{array} \right]$$

Time decay integration - $D(\Delta t) = e^{-\lambda \Delta t}$
[$\Delta t = t - t_{\text{last}}$] [λ = decay const. = $\ln 2 / T$] [T = half life]

$$W_{\text{total}} = \sqrt{B \times b} \times e^{-\frac{\lambda \Delta t}{2}}$$

neutrality w/ recency wt.

Balance evolution: $B' = B \cdot \frac{P_{\text{agg}}}{P_i}$; $b' = b \cdot \frac{P_i}{P_{\text{agg}}}$

$$U_i = E \left[\sum_{t=0}^{\infty} \delta^t (P_i - P_{\text{agg}}) (B_i - b_i) + \frac{W_i}{\sum W_j} \right]$$

NASH EQUILIBRIUM PROOF

1. The utility function :

Each user maximises U_i .

$$U_i = \mathbb{E} \left[\sum_{t=0}^{\infty} \delta^t \left(\underbrace{(p_i(t) - p_{\text{true}}(t)) \cdot (B_i(t) - b_i(t))}_{\text{Trading Profit}} + \underbrace{\frac{W_i(t)}{\sum_j W_j(t)} \cdot R(t)}_{\text{Reward Share}} \right) \right]$$

2. W is maximized at $B = b$:

For any fixed total capital $K=B+b$, the weight function

$$W = \sqrt{B \cdot b}$$

achieves its maximum when the allocation is balanced: $B=b=K/2$ by AM-GM inequality.

3. Equilibrium :

In the Karma protocol with weight function

$$W = B \cdot b \cdot e^{-\lambda \Delta t},$$

the unique Nash equilibrium strategy for all rational, sufficiently patient players ($\delta > \delta^*$) is:

- Position Strategy: Maintain balanced positions, $B_i \approx b_i$.
- Reporting Strategy: Submit unbiased reports, $p_i = p_{\text{true}} + \epsilon_i$, with $E[\epsilon_i] = 0$.

From this, any deviation from $B_i = b_i$ reduces W_i for fixed capital. A lower weight implies:

- Smaller immediate reward share,
- Reduced future influence,
- No compensating gain, since trading profit requires imbalance but imbalance itself reduces weight.

With $B_i = b_i$, the trading profit term vanishes because $(B_i - b_i) = 0$. Utility then depends only on reward share, which is determined by future weights.

Future balances evolve as:

$$B_i(t+1) = B_i(t) \cdot \frac{P_{\text{agg}}(t)}{P_{\text{true}}(t)}, \quad b_i(t+1) = b_i(t) \cdot \frac{2P_{\text{true}}(t) - P_{\text{agg}}(t)}{P_{\text{true}}(t)}.$$

This yields the next-period weight:

$$W_i(t+1) = B_i(t)b_i(t) \cdot \frac{P_{\text{agg}}(t)(2P_{\text{true}}(t) - P_{\text{agg}}(t))}{P_{\text{true}}(t)^2}.$$

The fraction is maximized when $P_{\text{agg}} = P_{\text{true}}$. Any bias in reporting reduces future weight, lowering long-run utility.

By the one-shot deviation principle, if no profitable deviation exists in a single period, the strategy is a subgame perfect equilibrium. When players are sufficiently patient ($\delta > \delta^*$), the discounted value of future losses from deviation outweighs any short-term gain.

Security Properties

Sybil Resistance :

$$\sum_{j=1}^k \sqrt{B_j b_j} \leq \sqrt{(\sum B_j)(\sum b_j)}$$

Proof: By Cauchy-Schwarz inequality, splitting never increases total weight.

Manipulation Resistance :

$$\left. \begin{aligned} \frac{\partial W_{\text{future}}}{\partial P_{\text{agg}}} &= 0 \\ \frac{\partial^2 W_{\text{future}}}{\partial P_{\text{agg}}^2} &< 0 \end{aligned} \right|_{P_{\text{agg}} = P_{\text{true}}}$$

(future weight maximized at truthful

reporting)

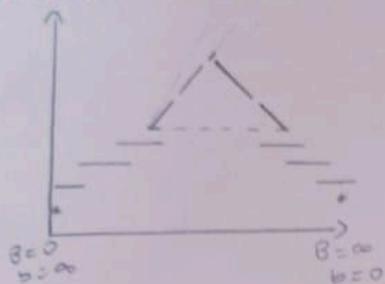
Flash Loan Resistance :

$$\lim_{\Delta t \rightarrow 0} W_{\text{total}} = 0$$

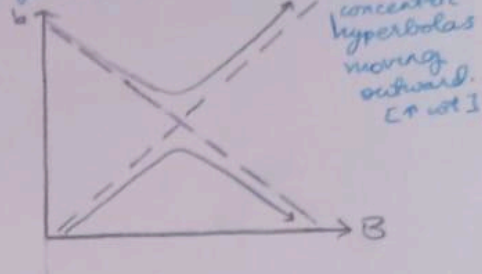
Collusion Resistance :

$$\sum_{i \in C} (B_i - b_i) \approx 0 \implies \sum_{i \in C} \pi_i^{\text{trade}} \approx 0$$

> 3D surface plot -
 $w = \sqrt{B \cdot b}$, $B, b \geq 0, w \in [0, \infty)$



> 2D Contour Plot (Top-Down View)
 for const. $w=c \rightarrow B \cdot b = c^2$



> $A+B=b \rightarrow w=\pi \rightarrow$ critical diagonal. 

> Hessian of $w = \sqrt{B \cdot b}$.

$$H = \begin{bmatrix} -k \cdot b/B & k \\ k & -k \cdot B/b \end{bmatrix} \text{ where } k = \frac{1}{4\sqrt{B \cdot b}}$$

$|H| = 0$. [singular $\rightarrow$ concave func. with one direction of zero curvature]

$\rightarrow$ eigen values $\rightarrow \lambda_1 = 0$, $\lambda_2 = -\frac{1}{4} \left(\frac{\sqrt{b}}{B^{3/2}} + \frac{\sqrt{B}}{b^{3/2}} \right) < 0$.

Thus Hessian is (-ve) semidefinite $\rightarrow$ concave func. but linear along scaling direction

1 Direction is flat (Scaling direction): $\lambda_1 = 0$; $V_0 = \begin{bmatrix} B \\ b \end{bmatrix}$ (neutral grows large)

Other direction is curved (Rebalancing): $\lambda_2 < 0$; $V_1 = \begin{bmatrix} b \\ -B \end{bmatrix}$ [stronger incentives as you are more unbalanced]

$$V^T H V = \frac{-1}{4\sqrt{B \cdot b}} \left[\sqrt{\frac{b}{B}} V_0 - \sqrt{\frac{B}{b}} V_1 \right]^2 \leq 0. \text{ [curve never +ve i.e. no tipping] pts is no increasing returns.}$$

There are no hidden convex regions where manipulation can create gains (proved by 2nd derivative + mixed partial derivatives)

For any fixed total capital $K=B+b$, the weight function

$$W=B \cdot b$$

achieves its maximum when the allocation is balanced: $B=b=K/2$.